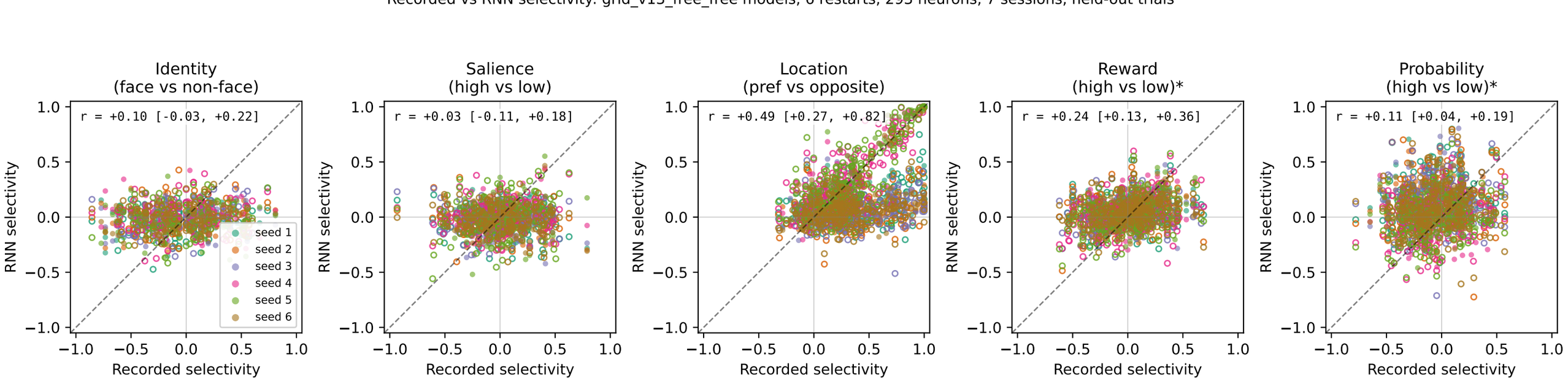

Recorded vs RNN selectivity: grid_v13_free_free models, 6 restarts, 293 neurons, 7 sessions, held-out trials
Identity
(face vs non-face)
r = +0.10 [-0.03, +0.22]
Salience
(high vs low)
r = +0.03 [-0.11, +0.18]
Location
(pref vs opposite)
r = +0.49 [+0.27, +0.82]
Reward
(high vs low)*
r = +0.24 [+0.13, +0.36]
Probability
(high vs low)*
r = +0.11 [+0.04, +0.19]
seed 1
seed 2
seed 3
seed 4
seed 5
seed 6
RNN selectivity
Recorded selectivity
1.0
0.5
0.0
−0.5
−1.0
Selectivity = 2 x (ROC area - 0.5), 50-250 ms after target onset, at each neuron's preferred location. Filled: E, open: I. r per seed in legend order (mean [min, max] for more than 3). RNN: training noise, Poisson spikes, mean of 10 draws.
* Trial reward and probability are not model inputs; context models receive the block reward map and probability context.